

SHADOWGRAM

Behavioral Graph Detection of Coordinated AI-Agent Swarms & Synthetic Persona Rings

TRACK 04: SYNTHETIC IDENTITY & KYC • TRACK 05: OPEN FRAUD

TEAM: TEAM SHADOWGRAM

INSTITUTION:

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What is being Faked/Exploited?

The Threat: Autonomous AI-Agentic Multi-Accounting & Coordinated Sybil Fraud Rings

1. The AI Exploit Scenario

- **Autonomous Multi-Accounting:**

Attackers deploy autonomous AI agents (Browser-Use, AutoGPT, LLM runners) that operate dozens of accounts in parallel.

- **Polymorphic Camouflage:** LLMs dynamically generate unique profile text, distinct bios, and conversational nuances, easily bypassing keyword filters.

- **Defeating Single Checks:** Traditional captchas and single-account rules fail because each synthetic persona behaves organically human in isolation.

2. Victims & Target Sectors

- **FinTech & Micro-Lending:** Coordinated referral abuse, sign-up bonus drainage, and automated micro-loan harvesting across multiple accounts.

- **E-Commerce & Marketplaces:** Distributed astroturfing, fake merchant syndicates, and wash-trading manipulation.

- **Digital Platforms:** Large-scale inauthentic sentiment manipulation, phishing distribution, and trust degradation across real ecosystems.

3. Why Existing Tools Fail

- **Opaque Black-Box Scores:** Current anti-fraud tools output unexplainable numbers (e.g., '82% Suspicious') without actionable evidence trails.

- **Isolated Session Silos:** Legacy tools inspect accounts one at a time, completely blind to cross-session relational coordination.

- **Unenforceable Bans:** Without inspectable causality, compliance teams cannot justify account suspensions, risking massive false-positive churn.

How will we stop it?

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Relational Behavioral Telemetry & The Explainable 'Why' Evidence Dossier

1. The Relational Shift

- **Core Thesis:** Shift from unanswerable 'Is this single account AI?' to tractable 'Are these independent accounts commanded by the same automated puppet master?'
- **Capturing Hidden Harmony:** While LLMs fake varied text, automated agent runners leave undeniable mathematical fingerprints across routing, timing, and keystrokes.
- **Multi-Signal Defense:** Requires convergence across 5 independent evidence layers before raising high-confidence flags.

2. The 5 Behavioral Fingerprints

- **Interaction Cadence:** Keystroke Flight Time (FT) and Dwell Time (DT) variance and pointer curvature jerk.
- **Navigation Sequences:** Finite State Machine (FSM) route transitions across application views (e.g., Home -> Search -> Post).
- **Micro-Temporal Sync:** Request ingress timestamp deltas ($\Delta t < 1.4s$) exposing automated batch submission bursts.
- **Semantic Embeddings:** all-MiniLM-L6-v2 dense vector cosine similarity (> 0.85) catching paraphrased spam.
- **Client Environment:** Canvas & WebGL hardware rendering hashes and device fingerprints.

3. The Explainable 'Why' Layer

- **The Centerpiece Feature:** Converts high-dimensional graph connections into plain-English, auditable evidence dossiers.
- **Inspectable Attribution:** 'Linked by 94% timing correlation, identical 4-step navigation route, and 0.89 semantic similarity.'
- **Scientifically Defensible:** Reports 'signals consistent with coordinated operation', avoiding unverifiable individual identity claims.

Implementation

100% Deterministic, Offline-Capable Systems Pipeline & 48-Hour MVP

1. Systems Pipeline & Stack

- **Client Telemetry SDK:** Lightweight unobtrusive JS event hooks capturing DOM events and route transitions in real time.
- **FastAPI Ingestion Engine:** High-throughput async ingestion normalizing raw telemetry into multi-variate feature vectors x in R^D .
- **NetworkX Graph Worker:** Evaluates multi-signal composite weights ($S_{comp} \geq 0.78$) and executes Louvain Community Detection to isolate fraud subgraphs.
- **Compliance Dashboard:** Next.js / Tailwind UI rendering real-time force-directed clusters and drill-down evidence cards.

2. Local Offline Execution

- **Zero Cloud API Reliance:** 100% independent of OpenAI/Claude or external cloud endpoints; immune to rate limits or hackathon Wi-Fi drops.
- **Sub-80ms Inference:** all-MiniLM-L6-v2 transformer (~90MB) executes CPU batch embeddings in $< 80ms$ on standard laptop hardware.
- **Minimal Resource Footprint:** Full graph clustering across $N=100$ accounts takes $< 120ms$ with $< 150MB$ RAM memory footprint.

3. 48h Demo Benchmark

- **Synthetic Evaluation Harness:** Simulates $N=100$ enterprise accounts: 70 Organic Users, 15 Synchronized Botnet Accounts, 10 Farm Users, 5 Outliers.
- **Mathematical Convergence:** Edge creation requires $P(\text{Convergence}) = \prod P(\text{Signal}_k)$ crossing strict statistical thresholds.
- **Benchmark Accuracy:** Empirically validated with F1-Score > 0.94 and False Positive Rate $< 2.5\%$, proving statistical rigor.

Conclusion

Transforming Opaque Risk Scores into Actionable Forensic Intelligence

1. Blast-Radius Containment

- **Uncontained Baseline:** A 15-node AI botnet remains active for weeks, harvesting Rs 4.5L+ in promotional bonuses and reaching 12,000+ users.
- **With ShadowGram:** Syndicate is unmasked upon the 3rd synchronized action; automated quarantine limits blast radius to < 100 users.
- **Proactive Containment:** Triggers step-up authentication (WebAuthn/FIDO2) or quarantine rather than blunt, disruptive account bans.

2. Drop-In Enterprise Fit

- **Lightweight Client SDK:** Embeds directly into banking web portals, fintech onboarding, or community platforms in < 10 lines of code.
- **REST & WebSocket Webhooks:** Exposes real-time alert webhooks for incident response (SIEM/SOAR integration).
- **Privacy & DPDP Compliance:** Client-side cryptographic telemetry hashing (no sensitive PII or password logging; DPDP Act compliant).

3. The Final Winning Takeaway

- **Beyond Black Boxes:** 'Most fraud systems output an unexplainable number. ShadowGram delivers an inspectable evidence trail built on multi-signal convergence.'
- **Judicial Trust:** Gives fraud analysts, compliance officers, and hackathon evaluators an auditable defense system they can verify and trust.
- **Demonstrable in 3 Minutes:** Complete end-to-end working loop ready for live physical adjudication at Jyothi Engineering College.